

melanoma

7670 TREATMENT OF PATIENTS WITH ADVANCED REFRACTORY OR RELAPSED MELANOMA IN A PHASE II STUDY OF TREMELIMUMAB (CP-675,206), AN ANTI-CTLA4 MONOCLONAL ANTIBODY

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The antitumor activity and safety of tremelimumab (CP-675,206), a fully human anti-CTLA4 monoclonal antibody, was assessed in a phase II, open-label, single-arm study in patients with advanced melanoma. Eligible patients had measurable, surgically incurable stage III or IV melanoma and had progressed, relapsed, or could not tolerate previous treatment. Tremelimumab (15 mg/kg) was administered intravenously every 12 weeks for up to 4 cycles (patients with clinical benefit were eligible for ≤ 4 additional doses) or until disease progression or intolerable toxicity. Proportion of confirmed objective RECIST responses per independent review was the primary endpoint, and secondary endpoints included safety, duration of response, and overall survival (OS). Among 251 enrolled patients, 246 patients (242 response-evaluable) received a median of 1 dose (range, 1 to 7). The objective response rate per RECIST, as determined by the Independent Endpoint Review Committee, was 6.6% (16 partial responses [PRs]), and all responses were durable (≥ 170 days). Eleven of these PRs are ongoing (6.0+ to 17.7+ months). The majority of PRs involved target lesions in lung, lymph nodes, and/or liver. An additional 36 patients had stable disease (SD; ≥ 70 days since first dose); therefore, the clinical benefit rate was 22% (PRs + SDs). Six patients (2.5%) had a PR in target lesions per RECIST while having overall progressive disease, and these patients remain alive on study with censored OS (14.82+ to 18.23+ months). Median OS for all patients was 10.1 months, and Kaplan-Meier estimate of 12-month survival was 41%. Most treatment-related adverse events (AEs) were mild to moderate. Grade 3 or 4 AEs included diarrhea (n = 28, 11.4%), fatigue (n = 6, 2.4%), and colitis (n = 6, 2.4%). There were 2 (0.8%) treatment-related deaths (1 sudden death and 1 diverticular perforation). While this study did not demonstrate a second-line response rate exceeding 10%, duration of response to this regimen suggests a role for tremelimumab in this population, and future assessment of predictive factors of response may serve to broaden understanding of clinical benefit observed in some patients.

7680 PROLONGED STABLE DISEASE IN IPILIMUMAB-TREATED PATIENTS WITH ADVANCED MELANOMA WHO HAVE PROGRESSED ON PRIOR ANTICANCER THERAPIES

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Objective: Ipilimumab is a fully human, monoclonal antibody directed against cytotoxic T lymphocyte antigen-4 (CTLA-4). Unique patterns of clinical response to ipilimumab are emerging from recent clinical trials. As more patients (pts) are treated with ipilimumab, more cases of prolonged stable disease (SD) with slowly declining tumor volume (≥25%), have been observed. This may be associated with the immunological mechanism of action of ipilimumab, which requires immune activation prior to a clinical antitumor effect and may take weeks to months to fully develop. Here, we summarize this clinical pattern of SD.

Methods: Ipilimumab was administered at 10 mg/kg every 3 weeks (Q3W) x 4 (induction dosing) in a single-arm, multicenter, phase II study in pts with advanced melanoma who had failed prior therapeutic regimen(s); eligible pts received maintenance Q12W starting at Week 24. The primary endpoint was the best overall response (complete + partial) rate. Follow-up tumor evaluation beyond progressive disease (PD) before the administration of non-ipilimumab anticancer therapy was permitted. Clinical observations of novel patterns of antitumor activity led to the definition of exploratory response categories, which were described as immune-related response criteria (RC). Immune-related RC consider measurable lesions (≥1cm) only, define total tumor burden as baseline index lesions + any measurable new lesions, and follow pts beyond traditional modified WHO PD to capture late activity.

Results: This analysis examined 155 pts. Based on immune-related RC, 22 pts (14.2%) were classified as having SD with slow ≥25% decline in tumor volume at last follow up. This SD pattern added meaningfully to immune-related complete and partial responses (50–100% reduction in tumor volume) for an overall rate of measurable antitumor activity of 25.8% (40 pts).

Conclusions: In pts treated with ipilimumab, SD was frequently characterized by a measurable tumor volume reduction of ≥25% from baseline. This clinical pattern of SD is likely a result of the clinical mechanism of action of ipilimumab and may contribute to the prolonged overall survival observed (results reported elsewhere).

7690 EFFECT OF DOSE ON EFFICACY AND SAFETY IN IPILIMUMAB-TREATED PATIENTS WITH ADVANCED MELANOMA – RESULTS FROM A PHASE II, RANDOMIZED, DOSE-RANGING STUDY

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Objective: Ipilimumab is a fully human monoclonal antibody directed against cytotoxic T-lymphocyte antigen-4 (CTLA-4). Here we report the effects of ipilimumab dose on response, survival, and safety in a recently completed, dose-ranging study of patients (pts) with previously treated, advanced melanoma. **Methods:** This was a randomized, double-blind, multicenter, phase II trial. Pts with unresectable stage III/IV melanoma who had progressive disease (PD) on, or were intolerant to, prior therapies received ipilimumab induction dosing at 0.3, 3, or 10 mg/kg every 3 weeks (Q3W) X 4. Eligible patients continued on maintenance therapy Q12W starting at Week 24. The primary endpoint was the best overall response rate (BORR) as determined by an Independent Review Committee using modified WHO criteria. The Kaplan-Meier product limit method was used to determine 1-year and overall survival (OS).

Results: After a median follow up of ~8 to 9 months, median OS and 1-year survival rate were numerically improved at 10 mg/kg compared with 3 and 0.3 mg/kg (Table). There was a statistically significant dose effect based on BORR (trend test P = 0.0015) with highest activity in the 10 mg/kg group. There was also an increase in immune-related adverse events from 0.3 to 3 to 10 mg/kg, suggesting a dose effect. Safety events were medically manageable, as observed in other studies.

Efficacy endpoint	10 mg/kg N = 72	3 mg/kg N = 72	0.3 mg/kg N = 73
OS — Median (95%CI), months	14.59 (6.9, —)	8.64 (6.87, 12.25)	8.57 (7.69, 14.49)
Survival follow-up — Median, (Interquartile Range), months	8.92 (3.24 - 12.07)	8.05 (3.91 - 11.48)	7.92 (3.29 - 10.78)
Survival rate at 1 year — %, 95%CI (%)	53.39 (41.17, 63.71)	38.16 (25.32, 50.94)	40.38 (27.06, 53.79)
BORR (complete + partial response [CR + PR] — N (%) - 95%CI (%)	8 (11.1) (4.9, 20.7)	3 (4.2) (0.9, 11.7)	0 (0) (0.0, 4.9)
Disease control rate (PR + CR + stable disease) — % (95% CI)	29.2 (19.0, 41.1)	26.4 (16.7, 38.1)	13.7 (6.8, 23.8)

Conclusions: The 10 mg/kg cohort had the highest median OS, survival rate at 1 year, and BORR. The survival outcome compares favorably with data reported in the literature¹. Together, these data suggest that 10 mg/kg ipilimumab offers a better benefit-risk ratio than the 3 and 0.3 mg/kg doses, and improves survival with an acceptable level of risk. 1. Korn et al. J Clin Oncol 26:527-34, 2008

7700 CELL CYCLE REGULATION IN MELANOMA: IMPLICATIONS OF CYCLIN D1 GENE VARIANTS WITH GENETIC SUSCEPTIBILITY TO MELANOMA

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Background: The rising incidence of melanoma and lack of effective treatments for advanced disease represent an important public health problem. Exposure to the sun is generally accepted as a major causative factor. However, its mechanism is unknown, and the role of exposure to ultraviolet light is complex. There is a growing body of evidence demonstrating that oncogenes are important contributors to the deregulation of cell proliferation and apoptotic pathways in melanoma. Cyclin D1 (CCND1) is a key regulator of the G1/S phase of the cell cycle and its altered activity is associated with the development of several human cancers, including melanoma. The elevation of CCND1 protein levels as a result of either mutations in upstream genes or increases in the gene dosage represents a crucial event driving progression in melanoma. Some studies have implicated CCND1 as an important oncogene in melanoma, making it candidate for further evaluation as a therapeutic target in this disease.

Material and methods: We analysed the A870G CCND1 polymorphism by PCR-RFLP in the genomic DNA isolated from peripheral blood of 1053 individuals, including, 161 cases with melanoma and 892 healthy individuals. Statistical analysis was performed using the computer software SPSS for Windows (version 11.5). Chi-square analysis was used to compare categorical variables and a 5% level of significance was used in the analysis. The odds ratio (OR) and its 95% confidence interval (CI) were calculated as a measurement of the association between CCND1 genotypes and melanoma risk. Logistic regression analysis was used to calculate the adjusted OR (aOR) and 95% CI for the influence of CCND1 genotypes in the risk of melanoma, with adjustment for age and gender.

Results: Our results indicate that individuals carrying two G-alleles have a 1.8-fold increase in the risk for the development of melanoma (OR=1.78, 95% CI 1.21-2.60, P=0.003). Multivariate logistic regression analysis confirmed the association between the presence of the GG CCND1 genotype and increased genetic susceptibility for the development of melanoma (aOR=1.97, 95% CI 1.27-3.05, P=0.002).

Conclusions: To the best of our knowledge, this is the first study suggesting that A870G CCND1 polymorphism is associated with the susceptibility to melanoma. Our results indicate that this polymorphism may be an important biomarker for melanoma risk, especially in the definition of a genetic predictive profile for the development of melanoma within our population.

7710 ELESCLOMOL (FORMERLY STA-4783) AND PACLITAXEL IN STAGE IV METASTATIC MELANOMA (MM): 2-YEAR OVERALL SURVIVAL (OS)

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Background: Elesclomol is a novel small-molecule oxidative stress inducer known to synergize with taxanes and certain other chemotherapeutics.

Methods: This 2:1 randomized, double-blinded, active-controlled multicenter study compared 213 mg/m² elesclomol co-infused with 80 mg/m² paclitaxel (E+P) versus paclitaxel alone (P). Patients (pts) received 4-week cycles of once weekly E+P or P for 3 weeks and 1 week's rest until disease progression. Eligible pts with Stage IV MM and 1 or 0 prior chemotherapies were evaluated every 8 weeks for progression by RECIST criteria. P-treated pts were eligible for post-progression crossover to E+P. This trial met

its primary endpoint, demonstrating a statistically significant improvement in progression-free survival. We now report the results of 2-year OS follow-up data.

Results: At 21 sites, a total of 81 subjects were randomized to E+P (53 pts) or P (28 pts). The median duration of study medication exposure was 102.0 d for E+P pts and 43.5 d for P pts. Of pts progressed on P, 19 crossed over to E+P. Median follow-up times for censored subjects exceeded 21 months for all subsets. Adverse events for E+P-treated pts were typical of P use, with slight/moderate incidence increases in E+P pts v. P pts for hypoesthesia (31% v. 4%), constipation (48% v. 29%), fatigue (69% v.50%), arthralgia (29% v. 11%), neutropenia (17% v. 0%), and stomatitis (15% v. 0%). Intent-to-Treat Analysis of Survival Time from Randomization

Group	E+P	P	P Crossover	P No Crossover
N	53	28	19	9
Died	42	22	14	8
Censored	11	6	5	1
Mo. Median OS (95% CI)	11.9 (8.5-17.7)	7.8 (5.7-16.0)	14.3 (6.9-20.5)	5.6 (2.3-6.1)
1-year-OS	49%	43%	53%	22%
2-year OS	27%	21%	26%	not estimable

Conclusions: The E+P safety profile was tolerable and similar to P alone. Crossover may have confounded OS comparison; nonetheless, encouraging median and 2-year OS results for E+P pts warrant further study of E+P in MM. A multinational Phase 3 trial of E+P versus P (SYMMETRYSM) is now enrolling.

772PD MIRNAS, GENE EXPRESSION AND ARRAY-CGH INTEGRATED ANALYSIS GIVES A NEW INSIGHT IN MELANOMA CANCER

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Introduction: In this study, we evaluated the results of an integrated analysis of melanoma. The scope of the study was to thoroughly investigate the effect of chromosomal aberrations on the alteration of specific miRNAs that regulate gene expression.

Materials and Methods: Twenty six human melanoma cell lines were obtained from M. Herlyn (Wistar Institute, Philadelphia, PA). Gene expression profiling was performed using Affymetrix HU-133A chips for 3 radial growth phase, 9 vertical growth phase, 10 metastatic melanoma as well as 4 melanocyte cell lines. results were analysed with the Microarray Processing Software to find all the differentially expressed genes. Furthermore, the chromosome regions for these genes were examined for possible aberrations with array CGH. Moreover, the assumption that these genes constitute miRNA targets was investigated in silico taking into account the possible precursor miRNA DNA areas found to be deleted.

Results: Twelve genes displayed altered expression in the different stages of the disease. Two of these were upregulated only in metastatic melanoma samples and were computationally found to be targeted by mir-136. It was then hypothesized that the precursor miRNA DNA region for mir-136 presented copy number alterations. Indeed, the analysis revealed a remarkably high frequency of mir-136 gene deletions in metastatic melanoma samples which is expected to be correlated with mir-136 expression levels. We claim that melanoma disease progression correlates with the loss of the corresponding miRNAs that regulate genes, which should be "switched off".

Conclusion: Our results suggest that mir-136 may play a crucial role in disease progression as a regulator of genes that are involved in the control of melanoma growth and invasion.

The findings support the notion that DNA copy number alterations of miRNAs not only correlate with the expression of their regulatory genes but may indeed play a major role in cancer progression. Possible therapeutic applications and the prognostic value of our findings need to be investigated in the near future.

773PD DETECTION OF DISTANT METASTASES WITH FDG-PET AND CT IN 251 CLINICALLY STAGE III MELANOMA PATIENTS

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Introduction. Distant metastases of melanoma can affect any organ or tissue and the presence of multiple metastases is usually a contraindication for surgery. Therefore an important part in the treatment planning is the evaluation of possible distant metastases. The aim of this prospective study was to assess the detection of distant metastases with FDG-PET and spiral CT in patients clinically stage III melanoma.

Methods. All consecutive patients with palpable, histology/cytology proven lymph node metastases of melanoma referred to the five participating hospitals between July 2003 and November 2007 for examination with FDG-PET and CT were prospectively included in this study. All patients were considered candidates for further surgical intervention in the form of a regional lymph node dissection. For each possible site of metastases the absence or presence of metastases was recorded.

Results. Overall, 251 patients were included in the five hospitals. More than half the patients were male, median age was 57.9 (range 23.2-93.1) years. The primary melanoma was mostly located on the trunk (37.0%) or lower extremities (35.0%). FDG-PET and CT were comparable in detecting liver metastases ($p=0.81$), lung metastases ($p=0.33$) and abdominal metastases ($p=0.62$). However, FDG-PET detected significantly more bone metastases ($p<0.0001$) and skin metastases ($p=0.03$). Overall, FDG-PET detected more metastatic sites as compared to CT ($p=0.017$). Treatment changed in 48 patients (19.1%), mostly as result of both scans (66.2%), or FDG-PET alone (20.8%) or CT alone (13.0%).

Conclusions. In conclusion, FDG-PET detected more metastatic sites than CT, especially bone and skin metastases. Treatment changed in 19% of the patients, mostly to chemotherapy. The PET scan, with the CT scan as a good alternative are indicated in staging of melanoma patients with palpable lymph node metastases.

774PD RANDOMIZED PHASE II STUDY OF DACARBAZINE (DTIC) WITH OR WITHOUT CARBOPLATIN (CBDCA) IN PATIENTS WITH STAGE IV CUTANEOUS MELANOMA

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Background: In the treatment of patients with advanced melanoma response rate with DTIC are now confirmed to be only 7.2-7.5 percent. Polychemotherapy regimens yielded higher response rates than monotherapy but without overall survival improvement. This randomized phase II study was performed to compare the efficacy and toxicity of DTIC vs DTIC-CBDCA in patients with advanced cutaneous melanoma.

Patients and methods: From 10/2000 to 10/2007 148 patients were enrolled (77 DTIC and DTIC-CBDCA 71); eligibility criteria included stage IV cutaneous melanoma, measurable disease, no prior cytotoxic chemotherapy, ECOG performance status 0-2. Patients were randomized to receive DTIC 220mg/m², day1-3, q 21 days (arm A) or DTIC 220mg/m², day 1-3 plus CBDCA AUC 5, day 1, q 21 days (arm B). Treatment arms were well balanced with regard to patient- and disease-related characteristics.

Results: By intent to treat analysis of 148 patients, the response rate was 11.69% (95%CI:6.27-20.75) for arm A vs 21.13% (95% CI:13.24-31.98) for arm B ($p=0.11$). The mean number of cycles was 4 (arm A) vs 5 (arm B) ($P=0.02$). Disease control rate (CR+PR+SD) was achieved in 33.77% in arm A vs 43.66% in arm B ($P=0.21$). The number of rapidly progressive disease after first cycle was higher in group A. The median follow-up was 6 months (range 1-47) for arm A vs 9 months (range 2-69) for arm B ($P=0.03$). The median PFS of DTIC vs DTIC-CBDCA was 6.5 months (95% CI: 5-10) vs. 8.0 months (95% CI: 7-12) ($P=0.14$). The median overall survival was 7 months (95% CI: 5-9) for arm A vs. 9 months (95% CI: 8-10) for arm B ($P=0.058$). The median overall survival for subgroup of patients with stage IVM1c was 5 months in arm A vs 8 months in arm B ($P=0.002$); HR=1.59(95% CI: 1.02-2.50). The three, six and nine months survival rates were for arm A -78%, 50% and 34% and for arm B - 89%, 70% and 42%. Toxicities were as follows: anemia gr.3 in 4 vs. 5 pts, granulocytopenia gr.3 in 4 vs. 6 pts, thrombocytopenia gr.3 in 1 vs. 4 pts, for arm A vs. arm B.

Conclusion: DTIC-CBDCA regimen was well tolerated, relatively cheap, with a trend towards better efficacy compared with DTIC, particularly for subgroup of patients with stage IVM1c, and not inferior in comparison with other frequently used drug combinations.

775PD A RANDOMISED, CONTROLLED STUDY TO EVALUATE THE EFFECT OF HIGH-DOSE BOSENTAN IN PATIENTS WITH STAGE IV METASTATIC MELANOMA BEING TREATED WITH DACARBAZINE

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Background: Evidence suggests that endothelin (ET) may play a role in the physiology of melanocytes and that the ET_B receptor is upregulated in human melanoma. The oral dual ET receptor antagonist bosentan, indicated for the treatment of pulmonary arterial hypertension, may potentiate the effects of alkylating agents. An open-label,

uncontrolled study suggested that bosentan may be of benefit in patients with stage IV metastatic melanoma. We therefore evaluated the potential benefit of bosentan combined with dacarbazine (DTIC) therapy in a phase II, randomised, double-blind, placebo-controlled trial.

Methods: This event-driven study included 80 patients with histologically proven malignant melanoma (stage IV, RECIST criteria), ECOG performance status ≤ 2 , no prior DTIC or other chemotherapy for stage IV disease, and no abnormal liver function tests or lactate dehydrogenase. Patients received intravenous DTIC 1000 mg/m² in 3-week cycles plus oral bosentan 500 mg bid or placebo. The primary endpoint was the time to tumour progression (TTP). Main analysis was on the all-randomised set.

Results: No difference between treatment groups was observed for TTP (hazard ratio [HR] placebo over bosentan: 1.14, 95% 2-sided confidence interval [CI] 0.72-1.83), progression-free survival (HR 1.06; 95% CI 0.66-1.70), or overall survival (HR 1.04; 95% CI 0.58-1.87). Incidences of most adverse events were similar between groups, as were clinically relevant ($\geq 3 \times$ upper limit of normal) increases in hepatic aminotransferases (10.5% placebo vs. 7.8% bosentan). A higher incidence of haemoglobin < 10 g/dL was observed with bosentan (5.2% placebo vs. 31.5% bosentan). This was managed by dose adjustment of DTIC; discontinuation of bosentan was not needed.

Conclusions: In this population of patients with severe stage IV metastatic melanoma, no treatment effect on time to tumour progression was observed with bosentan compared with placebo. No unexpected safety concerns or increased incidence of elevated hepatic aminotransferases were observed despite the high doses of bosentan used.

776PD EFFICACY AND SAFETY OF IPILIMUMAB IN PATIENTS WITH ADVANCED MELANOMA WHO HAD PROGRESSED ON ONE OR MORE PRIOR THERAPIES: RESULTS FROM A SINGLE-ARM, MULTICENTER STUDY

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Objective: The monoclonal antibody ipilimumab is specific for cytotoxic T-lymphocyte antigen-4 (CTLA-4), potentiating antitumor immune responses. In this analysis we report the effects of ipilimumab on the survival, efficacy, and tolerability of patients (pts) with advanced melanoma who developed progressive disease on after prior treatment (≥ 1 regimens).

Methods: In this single-arm, multicenter, phase II study, ipilimumab induction dosing therapy was administered at 10 mg/kg every 3 weeks (Q3W) X 4; eligible pts continued on ipilimumab maintenance therapy Q12W starting at Week (Wk) 24. The primary endpoint was best overall response rate (BORR). Tumor assessment (TA) was performed by an Independent Review Committee using a modified WHO (mWHO) criteria at Wk 12. The Kaplan-Meier product limit method was used to determine 1-year and overall survival (OS).

Results: Median follow-up for survival was 9.5 months. The median OS for the 155 treated pts was 10.22 months [95% CI 7.29, —]. While the BORR was only 5.8%, the survival rate at 1 year was 46.67% [95% bootstrap CI 38.60, 55.56]. Different patterns of clinical activity beyond conventional mWHO response, including responses after initial tumor volume increase, were observed and are reported elsewhere. Only 22% of pts experienced Grade 3/4 immune-related adverse events, which resolved with appropriate medical management. No bowel perforations were reported.

OS — median, months [95% CI]	10.22 [7.29, —]
Survival rate at 1 year — % [95% bootstrap CI]	46.67 [38.60, 55.56]
BORR (partial + complete response [PR + CR]) — N (%), [95%CI]	9/155 (5.8%), [2.7, 10.7]
Disease control rate (PR + CR + stable disease) — N (%), [95% CI]	42/155 (27.1), [20.3, 34.8]

Conclusions: In pts with advanced melanoma who had progressed on prior therapies and were treated with ipilimumab 10 mg/kg induction/maintenance dosing, the median OS compared favorably with data reported in the literature¹. While the response rate was 5.8%, several patterns of clinical activity unique to this drug class were observed (reported elsewhere) and may have contributed to the prolonged OS observed. Pts continue to be followed up. 1. Korn et al. J Clin Oncol 26:527-34, 2008.

777PD IMMUNE-RELATED ADVERSE EVENTS IN PATIENTS WITH METASTATIC MELANOMA TREATED WITH TREMELIMUMAB (CP-675,206)

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Tremelimumab (CP-675,206) is a fully human monoclonal antibody specific for cytotoxic T lymphocyte-associated antigen 4 (CTLA4) that prolongs and sustains T-cell activation and proliferation. Tremelimumab has demonstrated clinical antitumor activity in patients with advanced cancer and is well tolerated. Treatment with a CTLA4 blockade is associated with immune-related adverse events (IRAEs), as expected with enhancement of patient immunoreactivity. The incidence of IRAEs in the phase II portion of a phase I/II trial of tremelimumab in patients with metastatic melanoma was examined retrospectively. Patients with Eastern Cooperative Oncology Group performance status ≤ 1 who had progressed on or after prior therapy were treated with tremelimumab at 10 mg/kg monthly (n = 44) or 15 mg/kg every 3 months (n = 45). Similar numbers of patients in both cohorts experienced treatment-related adverse events of any grade (n = 31, 70% in the 10 mg/kg group; n = 30, 67% in the 15 mg/kg group), most of which were mild or moderate (grade 1 or 2) adverse events (n = 19/31, 61% in the 10 mg/kg group; n = 24/30, 80% in the 15 mg/kg group). Among patients with IRAEs, the first incidence of IRAEs generally occurred within the first cycle (58% in the 10 mg/kg monthly group; 100% in the 15 mg/kg quarterly cohort). The most common treatment-related IRAEs of any grade (eg, diarrhea, rash, and pruritus) were experienced with similar frequency in the 10 mg/kg and 15 mg/kg cohorts and were generally tolerable and reversible. Treatment-related diarrhea/colitis of $\geq$ grade 3 was reported by a higher number of patients in the 10 mg/kg cohort compared with the 15 mg/kg cohort (27% versus 11%, respectively). Consistent with the mechanism of action for an immune-enhancing drug, treatment with tremelimumab is associated with IRAEs. Unlike IRAEs observed with endogenous cytokines, most IRAEs observed with tremelimumab were tolerable and reversible, and some IRAEs were self-limited.

778PD EFFICACY AND SAFETY OF TREATMENT-NAÏVE AND PREVIOUSLY TREATED PATIENTS WITH ADVANCED MELANOMA RECEIVING IPILIMUMAB

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Objective: Ipilimumab is a fully human, anticytotoxic T-lymphocyte antigen-4 (CTLA-4) monoclonal antibody. This exploratory analysis examined the efficacy and safety of patients (pts) with advanced melanoma treated with ipilimumab who were treatment-naïve or had received prior systemic anticancer therapy.

Methods: This was a recently completed, randomized, double-blind, placebo-controlled, multicenter, phase II study. Ipilimumab $\pm$ prophylactic budesonide (PB) or placebo was administered to all patients (treatment-naïve or previously treated pts) with unresectable stage III/IV melanoma. The primary endpoint of this study was the rate of Grade ≥ 2 diarrhea. Budesonide was administered prophylactically to determine if diarrhea (an immune-related adverse event associated with ipilimumab therapy) could be prevented. Pts received ipilimumab induction dosing at 10 mg/kg every 3 weeks (Q3W) X 4; eligible pts continued with maintenance ipilimumab dosing Q12W starting at Week 24. Response was assessed using modified WHO criteria by an Independent Review Committee. Overall survival (OS) was assessed using the Kaplan-Meier product limit method.

Results: There were 53 treatment-naïve and 62 previously treated pts. To date, the median OS has not been reached in pts who were treatment-naïve (Table). The best overall response rate (BORR, Table) and the rate of Grade ≥ 2 diarrhea (results reported elsewhere) were also similar between groups.

	Treatment-naïve n = 53		Previously treated (n = 62)	
	Placebo (n = 32)	Budesonide (n = 21)	Placebo (n = 25)	Budesonide (n = 37)
OS — median, months (95% CI)	Not reached (11.99, —)	Not reached (—, —)	11.53 (6.64, —)	15.08 (6.47, —)
Survival rate at 1 year — % (95% CI)	67.22 (44.74, 83.45)	71.11 (48.48, 85.71)	48.79 (27.00, 68.42)	51.40 (35.14, 67.75)
Survival follow up — median, months	11.24	11.10	9.89	7.29
BORR (CR + PR) % (95% CI)	15.6 (5.3 – 32.8)	9.5 (1.2 – 30.4)	16.0 (4.5 – 36.1)	13.5 (4.5 – 28.8)

Conclusions: Survival of pts with advanced melanoma treated with ipilimumab, who were either treatment-naïve or previously treated, compares favorably with data reported in the literature¹. Survival may be longer in treatment-naïve pts. A phase III study of 500 pts with advanced melanoma is ongoing to determine the survival of pts receiving ipilimumab plus dacarbazine in treatment-naïve pts. 1. Korn et al. J Clin Oncol 26:527-34, 2008.

779P URUGUAYAN EXPERIENCE IN UVEAL MELANOMA: RESULTS ON 72 PATIENTS FOLLOWED DURING A 23-YEAR PERIOD

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Background: Uveal melanoma is the most common primary malignant intraocular tumor in adults (70%). They generally occur sporadically and family uveal melanoma only accounts for 0.6 % of patients.

Methods: Data was collected from 72 patients assisted at six different institutions in Montevideo, Uruguay, during a 23 year period. Diagnosis was histopathologically confirmed in 64/72 (88%) patients. Radical surgery (predominately enucleation) was performed in 60/72 (83%) patients. Other treatment options included orbit exenteration, brachitherapy, and proton beam therapy. 28/72 (38%) developed distant metastasis, mainly in the liver.

Results: The mean age at diagnosis was 59 years old. The most common symptom was decreased visual acuity in 58/72 (80%). Three patients had extrascleral extension of the tumor and presented with pigmented lesions. There was a statistically significant correlation between tumors of mixed and epithelioid type and survival (P<0.01) but there was no correlation between tumor diameter and survival (P=0.06). There were three clinical cases of familial uveal melanoma involving first generation members of the same family (siblings) one of which also had a malignant breast tumor diagnosed 2 years before the ocular tumor. A BRCA2 mutation was then identified. The three were enucleated and died with a mean time of 19 months because of distant metastasis. Secondary metastasis in the liver was imagenology confirmed in 26/72 (36%) patients. The median time from the diagnosis of the ocular tumor to the metastasis was 40.5 months and the median survival for the metastatic disease was 4 months. The median overall survival was 88 months (C.I. 95% 75-100).

Conclusions: This abstract illustrates the natural history of this entity taking into account the heterogeneous behaviour of it in our country, mainly with the occurrence of the extremely rare and fatal familial uveal melanoma (three siblings), and the extrascleral compromise as the presenting feature of three other cases.

780P THE ROLE OF PET SCANNING AND SENTINEL NODE BIOPSY IN MELANOMA: THE PATIENTS' PERSPECTIVE

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Background: The clinical role of Positron Emission Tomography (PET) and Sentinel Node Biopsy (SNB) in the management of patients with melanoma is becoming defined. However, little is known about the patients' views of the utility of these investigations. Aim: To investigate what patients with melanoma perceive to be the outcome and the significance of the PET scan and the SNB.

Patients and methods: One hundred patients with melanoma attending the outpatient clinic were eligible. Their views were elicited using a semi structured questionnaire. Descriptive statistical methods were used for the analysis of the participants' responses.

Results: Demographics: median age 55.7 years (SD 16.529), 54% were women, 66% had a higher level of education. The majority of patients were free of disease (82%).

PET scans: Only one third of the responders felt the doctor should decide if and when they should have a PET scan as part of their management. Fifty percent (50%) stated they wanted this test regularly (e.g every year) not only for awareness but also for reassurance. About 7% preferred to have it once upon diagnosis and a further 8% only if they had symptoms. SNB: Only about one third (38%) of those who had undergone a SNB had appreciated its prognostic significance whereas half of the patients (48%) followed their doctor's suggestion. Older patients were less keen to be involved in the decision making compared to younger patients ($p=0.017$). Patients identified the specialist who asked for their consent as a surgeon in 53 %, a dermatologist in 32 % and an oncologist in 7% of cases and 8% did not know who they gave consent to. Although the majority of patients were correctly aware of the significance of the SNB result, 4% of patients did not understand the results, 5% stated that they were never given the results and 14% of the responders thought they had the procedure to be cured.

Conclusion: Patients with melanoma want to receive information on investigations. Regular PET scans provide reassurance. The role and outcome of the SNB is not clear to all patients. More efforts should be made towards ensuring that patients are involved in the decision making process. Incorporating patients' views is essential in developing services in melanoma.

781P EVALUATION OF SERUM S-100B PROTEIN, 5-S-CYSTEINYLDOPA AND LDH IN PATIENTS WITH METASTATIC MELANOMA

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Objective: In the recent literature LDH was defined as the most important serum parameter in disseminated melanoma. S-100B protein, an acidic, calcium binding, cytoplasmic protein of low molecular weight is widely used as a serum marker and a risk factor in melanoma. 5-S-cysteinyldopa is a precursor of melanin. Purpose of the study was to compare the prognostic values of serum S-100B, LDH and 5-S-Cysteinyldopa concentrations in patients with metastatic melanoma.

Method: Serum samples were taken from 253 Stage IV metastatic melanoma patients (with hepatic, lung, brain and other metastases) at diagnosis. Serum S-5-CD concentrations were determined by HPLC, S-100B protein by immunoluminometric assay, while LDH was analysed conventionally, by UV kinetic method. Metastases were verified by histology or radiographic procedures (chest X-ray, abdominal ultrasound, MRI, CT). Significant differences were calculated by Mann Whitney method, survival of patients with normal and elevated serum marker level was illustrated by Kaplan Meyer survival curve and analyzed by Mantel Cox test.

Results and Conclusions: Mean marker concentrations of patients with progressive disease differed significantly from nonprogressive cases. Survival analysis indicated that the initially elevated marker levels in Stage IV disease predict comparably shorter survival. Results of our study suggest that these serum marker values correlate well with Stages, with tumor load and disease progression. The markers had appropriate sensitivity, high specificity as well as important positive predictive value. Among the studied markers S-100B protein proved to be the most reliable marker in respect to the clinical outcome.

782P SHARED REGIONS OF DECREASED DNA COPY NUMBERS IN NRAS AND BRAF MUTATED MELANOMAS

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Background: The NRAS and BRAF genes are commonly mutated in melanomas with reported frequencies between 5-36% and 25-80%, respectively. Decreased DNA copy numbers are found in melanoma cell lines with reported homozygous deletions covering chromosomes 9p21 and 10q23 regions (Oncogene. 2007 Jul 12;26(32):4738-48). Since limited information is available regarding decreased DNA copy numbers in melanomas with defined protooncogene mutations, we carried out a study aiming to identify those regions in NRAS and BRAF mutated metastatic melanomas.

Methods: A cohort of 21 metastatic melanoma samples that harbored either NRAS or BRAF mutations were used in the study. DNA from the fresh frozen samples was isolated and subsequently hybridized on Affymetrix 250K chips (Affymetrix, Santa Clara, CA, USA). SNP signal quantification were obtained with "GeneChip Operating

System" - GCOS (Affymetrix) and "GeneChip genotyping software" - GTYPE (Affymetrix) with call rate of 90% being used as a marker for good-quality array hybridization. Additionally, gene copy number dataset from 32 HapMap samples/ individuals was used as a reference for our data.

Results: Common regions in NRAS and BRAF mutated melanomas that had 1 gene copy were: 5q34, 14q23.1 and 19p13.2. While decrease in DNA copy numbers of the chromosome 10 (10p14, 10p13, 10p12.1 and 10q24.32) were found only in BRAF mutated tumors, 9p22.3, 9p11.1 and 11p15.5 regions were specifically linked to reduction of DNA copy numbers in tumors harboring NRAS mutation.

Conclusion: Metastatic melanomas harboring either NRAS or BRAF mutation have several shared regions of decreased DNA copy numbers, possibly indicating the presence of significant tumor suppressor genes in these regions.

783P EFFICACY AND SAFETY OF PATIENTS WITH ADVANCED MELANOMA TREATED WITH IPILIMUMAB WITH OR WITHOUT THE ADDITION OF PROPHYLACTIC BUDESONIDE

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Objective: Survival data are presented from a randomized, double-blind, multicenter, phase II study are reported. The effects of prophylactic budesonide (PB), an oral steroid with minimal systemic exposure used to treat inflammatory bowel disease, on the clinical activity of ipilimumab, a monoclonal antibody directed against cytotoxic T-lymphocyte antigen-4 (CTLA-4) were investigated.

Methods: Ipilimumab was administered to both treatment-naïve and patients (pts) who had received prior systemic therapy with stage III (unresectable)/IV melanoma at an induction dose of 10 mg/kg every 3 weeks (Q3W) x 4, concurrently with placebo (Group A) or PB QD at 9 mg to Week (Wk) 12, 6 mg to Wk 14, and 3 mg to Wk 16 (Group B). Ipilimumab maintenance therapy was offered to eligible pts Q12W at Wk 24. Response was assessed by an Independent Review Committee using a modified WHO criteria (mWHO). The Kaplan-Meier product limit method was used to determine 1-year and overall survival (OS). The primary endpoint of this study was the rate of Grade ≥ 2 diarrhea.

Results: 57 and 58 pts were treated in Groups A and B, respectively (115 total). PB did not have any impact on response or 1-year survival rate. Median OS was not yet reached for either group. Approximately 19% of pts per group had SD per mWHO. Immune-related adverse events were largely low grade and generally medically manageable; no bowel perforations were observed.

	Group A (Ipilimumab + placebo) N = 57	Group B (Ipilimumab + PB) N = 58
Grade ≥ 2 diarrhea — % [95% CI]	35.1 [22.9, 48.9]	32.8% [21.0, 46.3]
OS — median, months [95% CI]	Not reached [11.53, —]	Not reached [6.8, —]
Survival rate at 1 year — % [95% CI]	59.11 [45.38, 72.22]	58.79 [43.64, 70.34]
Survival follow-up — median, months	10.87	10.55
Best overall response (CR + PR) — N (%)	9 (15.8)	7 (12.1)
Disease control rate (PR + CR + SD) — % [95% CI]	35.1 [22.9, 48.9]	31.0 [19.5 – 44.5]

Conclusions: The median overall and 1-year survival rates of pts with advanced melanoma treated with ipilimumab compare favorably with data reported in the literature¹. PB did not appear to have a clinically meaningful effect on prophylaxis of diarrhea or 1-year survival. Further survival follow-up is ongoing. 1. Korn et al. J Clin Oncol 26:527-34, 2008

784P IPILIMUMAB-MEDIATED PATTERNS OF RESPONSE IN PATIENTS WITH PRETREATED, ADVANCED MELANOMA

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Objective: Ipilimumab, an anti-cytotoxic T lymphocyte antigen-4 (CTLA-4) monoclonal antibody, is a member of a novel class of immunotherapeutics. Measurable antitumor activity to ipilimumab may be observed after several weeks/months of treatment, both before and after progressive disease (PD). Although modified World Health Organization (mWHO) response criteria capture immediate responses, they do not capture responses that are late (after PD) or occur in the presence of new lesions seen with ipilimumab. More broad assessment criteria are needed to ensure that response to this novel class of immunotherapy can be recorded. This may help ensure that patients (pts) with PD by mWHO who may demonstrate benefit at a subsequent near term timepoint are not denied further treatment or follow-up.

Methods: Ipilimumab was administered at 10 mg/kg every 3 weeks (Q3W) x 4 (induction dosing) in a single-arm, multicenter, phase II study (CA184-008). Pts with advanced melanoma had experienced PD during/after prior therapeutic regimen(s). The primary endpoint was the best overall response rate (BORR). Eligible pts received maintenance ipilimumab Q12W starting at Week 24. Activity was assessed using mWHO and novel immune-related response criteria (RC) using previously defined mWHO thresholds for response. Contrary to mWHO, immune-related RC considered measurable lesions (≥ 1 cm) only; defined total tumor burden (TTB) as baseline index lesions + new lesions; and followed pts beyond mWHO PD to capture late activity. Here, we compare the BORR per mWHO with that of immune-related RC.

Results: For 155 pts treated, the mWHO BORR was 5.8%. The immune-related RC BORR was 9%. Late responses or SD with a decline in tumor volume $\geq 25\%$ from baseline were also observed. Together, these response patterns contributed to the totality of pts with an objective TTB reduction from baseline (25.8%).

Conclusions: Immune-related RC offer a systematic approach to capture novel patterns of response which 1) may be related to the immunopotentiating mechanism of action of immunotherapies such as ipilimumab and 2) may contribute to survival improvements (results reported elsewhere). Immune-related RC require prospective validation and/or further refinement.

785P IPILIMUMAB 10 MG/KG INDUCTION DOSING PROMOTES T-CELL ACTIVATION IN PATIENTS WITH ADVANCED MELANOMA

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Background: Ipilimumab is a fully human, monoclonal antibody targeting cytotoxic T-lymphocyte antigen-4 (CTLA-4). Its mechanism of action involves enhancing global T-cell immunity.

Methods: This was a randomized, double-blind, placebo-controlled, multicenter, phase II study. Patients (Pts) with stage III (unresectable) or IV melanoma received ipilimumab induction dosing 10 mg/kg every 3 weeks (Q3W) X 4, starting on Week (Wk) 1 and ending on Wk 12. Pts also received 15 weeks of prophylactic oral budesonide (PB) or placebo. The study objective was to investigate the effects of PB on rate of $\geq$ Grade 2 diarrhea (results reported elsewhere). In addition, pre- and post-treatment blood and serum samples were collected to assess CD4+ and CD8+ T-cell populations by flow cytometric analysis including activated (HLA-DR+ or CD69+) and naive (CD45RA+CCR7+) T cells.

Results: 115 pts were treated with ipilimumab (+ placebo, N = 57; + PB, N = 58). Across all evaluable pts, an overall mean increase of $\sim 10\%$ in activated CD4+ and CD8+ T cells and decrease of $\sim 6\%$ in naive CD4+ and CD8+ T cells was observed by Wk 4 (after the second dose). By Wk 4, $\sim 96\%$ of evaluable pts had an increase in activated T-cells and $\sim 82\%$ of evaluable pts had a decrease in naive T cells (Table). On an individual pt level, an increase in T-cell activation markers was associated with a concurrent reduction in naive T cells. In $>50\%$ of evaluable pts, these changes continued to evolve between Wks 4 and 12. An overall increase in the mean percentage of both central memory CD4+ (range 5–8%) and CD8+ (range 3–6%) T cells was observed at Wks 4 and 12 in all evaluable pts.

Activated T Cells	Frequency of patients with increase n/N (%)		
	Baseline to Week 4	Baseline to Week 12	Weeks 4 to Week 12
Activated CD4 T cells/Total CD4 T cells	49/51 (96.1)	32/33 (97)	10/30 (33.3)
Activated CD8 T cells/Total CD8 T cells	49/51 (96.1)	32/33 (97)	16/28 (57.1)
Naive T Cells	Frequency of patients with decrease n/N (%)		
	Baseline to Week 4	Baseline to Week 12	Weeks 4 to Week 12
Naive CD4 T Cells/Total CD4 T cells	24/28 (85.7)	20/22 (90.9)	9/15 (60)
Naive CD8 T cells/Total CD8 T cells	23/28 (82.1)	17/21 (81)	10/14 (71.4)

Conclusions: Flow cytometry data demonstrate that the induction dosing schedule of ipilimumab at 10 mg/kg Q3W X 4 activates T cells in the majority of evaluable pts. Association between a change in these T-cell parameters and clinical outcomes will be evaluated. The increase in activated T cells is supportive of the T-cell stimulatory mechanism of action of ipilimumab.

786P CLINICAL ACTIVITY OF IPILIMUMAB IN PATIENTS WITH ADVANCED MELANOMA AND BRAIN METASTASES

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Background: Ipilimumab, an anti-cytotoxic T-lymphocyte antigen-4 (CTLA-4) monoclonal antibody, showed clinical activity in phase I/II trials of patients (pts) with advanced melanoma. Anecdotal evidence suggests that ipilimumab may induce antitumor T-cell activity against brain metastases.

Methods: In this randomized, double-blind, multicenter, phase II study, pts with unresectable stage III or IV melanoma received ipilimumab 10 mg/kg every 3 weeks (Q3W) x 4 induction dosing in combination with placebo or prophylactic budesonide. Pts were either treatment-naïve or had been previously treated. Stable brain metastases at baseline were allowed, as the study primary endpoint was safety (results reported elsewhere). Eligible pts received maintenance ipilimumab Q12W starting at Week 24. The effect of ipilimumab in pts with brain metastases was assessed. Best overall response (BOR) was determined by an Independent Review Committee (IRC) using modified WHO criteria. The first tumor assessment was at Week 12.

Results: Of 115 total pts, 12 had brain metastases at baseline per the IRC. Of these, the BOR was partial response in 1 pt, unknown in 2, and stable disease in 3. In the only pt with a brain lesion as an index lesion (34.9 X 23.3 mm), the lesion decreased by Week 12 (classified as “too small to measure”). Based on the initial survival analysis at 6 months, 1 pt survived less than 6 months (died at 2.69 months) and 3 survived between 6 and 9 months (died after 6.64, 6.8, and 9.26 months). The remaining 8 pts were still alive at: 2.56+, 5.32+, 6.05+, 6.11+, 6.74+, 8.51+, 10.81+, and 14.06+ months. Four additional pts with baseline brain metastases per an individual IRC reviewer are included in the safety evaluation. Adverse events (AEs) involving the central nervous system were reported for 5 of these 16 pts, and considered drug-related in 2 (2 with Grade [Gr] 2 headache and 1 with Gr 1 dizziness). Two had Gr ≥ 3 AEs involving the nervous system, both reported as unrelated to ipilimumab: 1 with Gr 3 brain edema (Day 29) and 1 with Gr 4 seizure (Day 10).

Conclusions: A small number of melanoma pts with brain metastases treated with ipilimumab experienced extended, ongoing survival benefit. A study in advanced melanoma pts with brain metastases is ongoing.

787P TREATMENT GUIDELINES FOR THE MANAGEMENT OF IMMUNE-RELATED ADVERSE EVENTS IN PATIENTS TREATED WITH IPILIMUMAB, AN ANTI-CTLA4 THERAPY

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Background: Ipilimumab is a fully human anti-cytotoxic T lymphocyte antigen-4 (CTLA-4) monoclonal antibody being developed for patients (pts) with advanced melanoma and other malignancies. Adverse events (AEs) associated with ipilimumab are immune-related and typically affect gastrointestinal (GI), skin, liver, and

endocrine organs. Physician attention to and prompt treatment of immune-related AEs generally results in their resolution without apparent interference with antitumor efficacy.

Methods: Based on experience from early ipilimumab studies, specific treatment guidelines were implemented for immune-related AEs involving GI, liver, and endocrine organs. Physicians and pts were trained about the signs/symptoms of immune-related AEs to facilitate prompt detection and treatment. Suspected immune-related AEs were first evaluated to rule out non-immune-related etiologies. High- vs. low-grade AEs were treated differentially; low-grade events were treated mostly symptomatically, while high-grade events were treated with high-dose IV corticosteroids, tapered over 4–6 weeks. Persistent or relapsing low-grade GI or liver immune-related AEs were managed similar to high-grade immune-related AEs. Rare steroid-refractory, high-grade immune-related AEs were treated with secondary immunosuppressive regimens specific to each type. Occurrence of high-grade events precluded additional therapy with ipilimumab, with the exception of endocrinopathies managed with hormone-replacement therapy.

Results: The majority of immune-related AEs were low grade. High-grade events occurred in approximately 1/4 of pts of which only a small minority required secondary immunosuppressive therapy. Using established treatment guidelines, high-grade immune-related AEs were generally reversible and medically manageable on an outpatient basis in most pts by either withholding ipilimumab therapy or administering symptomatic therapy or corticosteroids. Management guidelines will be presented.

Conclusions: Treatment guidelines appeared to optimize pt safety and preserve ipilimumab efficacy. For high-grade events, management with corticosteroids is required.

788P ITALIAN MULTICENTER PHASE II TRIAL USING FOTEMUSTINE PLUS BEVACIZUMAB AS FIRST-LINE THERAPY IN METASTATIC MELANOMA

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Background: Fotemustine induces a 20–25% response rate (RR) in previously untreated metastatic patients (pts). Bevacizumab is an anti-VEGF monoclonal antibody with proven activity in metastatic colorectal, breast and non small cell lung cancer. Our primary study endpoint consisted in demonstrating an additive/synergistic activity of the fotemustine-bevacizumab combination in previously untreated metastatic melanoma pts.

Patients and methods: This is a classic phase II trial designed according to Simon two-stage hypothesis. 20 pts entered into the study, 18 of whom included in the interim analysis. They were treated with up to 6 cycles of fotemustine 100 mg/m² days 1,8,15 and every 3 weeks at the same dosage after 4 weeks of rest, combined with bevacizumab 15 mg/kg day 1 every 21 days.

Results: We observed 2 partial remissions and 10 stabilizations of disease (SD). The median duration of stabilization was 8 months (range 6–11+). Three SD are ongoing (10+,11+,11+). The disease sites in the two responsive pts were superficial lymph nodes (M1a stage). Among the pts with SD, three had also liver metastases. The median time to progression was 7 months (range 2–11+). The median survival was 8 months (range 2–13+). The clinical component of the trial was integrated with some biological evaluations, consisting in the determination of the soluble VEGFR (ELISA) and the pro-angiogenic and anti-angiogenic balance in the patients' sera. The results of such tests are not currently available but will be ready shortly.

Conclusion: The study was closed after the first stage for not having obtained the requested number of responses and for the incidence of grade 3–4 side effects (more than 40%), particularly in terms of myelotoxicity. However, the observed SD in a meaningful number of pts (10/18) could induce to hypothesize the use of bevacizumab either in combination with stronger/more effective chemotherapeutic regimens or as maintenance following induction biochemotherapy able to significantly reduce the tumor burden. The authors would like to thank the scientific service of the I.T.M.O. for its editorial assistance

789P ADJUVANT BEVACIZUMAB IS SAFE TREATMENT FOR MELANOMA PATIENTS AT HIGH RISK OF RECURRENCE

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Adjuvant therapy after resection of high risk melanoma, has, to date, not been shown to convincingly improve overall survival. Bevacizumab (Avastin®) is a recombinant humanised monoclonal antibody to vascular endothelial growth factor (VEGF), shown to improve survival in several incurable cancers. Biological studies implicate the VEGF pathway as being a relevant target in melanoma, while several phase II studies in advanced disease suggest bevacizumab has activity in melanoma. AVAST-M is a randomised trial evaluating bevacizumab as adjuvant therapy following resection of AJCC stage IIB (T4aN0M0), IIC (T4bN0M0) and III (TxN1-3M0) cutaneous melanoma. This CRUK-funded multicentre trial opened to recruitment in the UK in July 2007. Eligible patients are being randomised to receive bevacizumab (7.5 mg/kg IV infusion every 3 weeks), administered for 1 year, or to standard observation. Overall survival will be compared in 1320 patients, to identify an 8% difference in overall survival. Relapse-free and distant-metastasis-free survival, quality of life, toxicity and biomarkers are secondary end points.

Toxicity associated with bevacizumab is now well-described, but mostly relates to patients with advanced cancers receiving concomitant chemotherapy. We undertook a safety review of the first year of recruitment.

As of April 1st 2008, 71 patients had been recruited: 37 treatment, 34 observation. Mean duration of treatment is 3.1 months (range 0.5–8.4 months). To date, 5 pts have withdrawn, 4 due to distant recurrence, 1 due to toxicity. There have been no patient deaths.

Two SAEs have been reported. A patient on observation experienced a grade 3 skin infection and was hospitalised. A 48 year old man noticed right eye aching and loss of vision on day 11 after a first cycle of bevacizumab. By day 18 he had lost all sight in that eye. Ophthalmic review on day 19 suggested optic neuritis. A CT head scan was normal, but an MRI scan suggested ischaemic changes. This SAE has been reported as a SUSAR, the patient was withdrawn from treatment, has regained some vision and is on follow-up.

A 9 month safety review of the AVAST-M trial suggests adjuvant bevacizumab is very well tolerated. This report will be updated to include the first 12 months of recruitment.

790P A MULTICENTRE, PHASE II STUDY OF CP-4055 IN COMBINATION WITH SORAFENIB IN PATIENTS WITH METASTATIC MALIGNANT MELANOMA

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Background: CP-4055 (5'-elaidic acid ester of cytarabine) is a novel chemical entity developed as an antineoplastic agent with observed preclinical cytotoxicity in solid tumours. Promising early phase clinical results with CP-4055 in malignant melanoma (1 PR and several long-lasting SDs) have led to the present study. The aim is to evaluate the antitumor effect and the safety profile of CP-4055 given in combination with sorafenib (Nexavar®), an oral multiple kinase inhibitor, in malignant melanoma patients (pts).

Methods: CP-4055 is administered at days (d) 1–5 every 4 weeks (q4w) in a 30 min IV infusion. Starting dose of CP-4055 was 150 mg/m²/d (3 pts) and is continuing at 200 mg/m²/d. Sorafenib is given orally every day at 400 mg b.i.d. (800 mg/d). Activity is assessed at the end of each 2nd cycle (cy). If one PR or better is observed among the first 18 pts (step 1), 22 additional pts will be recruited (step 2).

Results: 17 pts have been enrolled, eight are ongoing and data is still under collection. Meaningful stabilisation of disease has been seen in two patients, including a 25% reduction in tumour size in one patient. Fatigue, anorexia, vomiting, fever, skin rash and nausea are the most observed adverse events. No significant unexpected AEs have occurred, but dose reduction of both drugs has been necessary following some cases of adverse skin reactions.

Conclusions: Infusion of CP-4055 (200 mg/m²/d) in combination with sorafenib 400 b.i.d. is generally well tolerated by patients with malignant melanoma in a d1-5 q4w schedule. However, the combination of the drugs has in some cases led to toxicity induced dose reduction. Recruitment is ongoing and updated efficacy and safety results will be presented.

791P OPEN-LABEL PHASE II STUDY OF SORAFENIB + DACARBAZINE IN PATIENTS WITH ADVANCED METASTATIC MELANOMA

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Background: Sorafenib (SOR), a potent and selective multi-kinase inhibitor, exerts its anti-tumor and anti-angiogenic effects via inhibition of VEGFR-1, -2, -3, PDGFR- α , - β , and Dacarbazine (DTIC) is a FDA-approved cytotoxic agent for advanced melanoma. Phase I/II results of SOR+DTIC showed encouraging activity, which warranted this open-label phase II study.

Methods: This was a multi-centre, open-label, uncontrolled Phase II trial to evaluate the primary endpoints of efficacy (disease control rate (DCR): CR+PR+SD) and tolerability of SOR+DTIC in patients with advanced metastatic melanoma. Eligibility criteria included measurable disease by RECIST, stage IV, and ECOG 0, 1 or 2. Oral sorafenib 400 mg twice daily (bid) was administered up to 6 cycles of 21 days with DTIC 1,000 mg/m².

Results: 24 patients with metastatic melanoma (mean age 56.4 yrs, SD=14.3; 54.2% men; ECOG 0-1) were treated with SOR+DTIC. All of them, 24 patients (100.0%), were considered evaluable for safety analysis, and 22 (91.7%) for efficacy analysis, after excluding 2 patients (8.3%) because of non-data for any primary endpoint of efficacy. Overall, 10 patients (45.5%) reached DCR (0 CR + 2 PR + 8 SD) as best response. Median progression-free survival was 2.2 months (95% CI: 0.2–4.1) (range: 0.5 – 6.13 months). Toxicity was assessed for all patients. 181 adverse events were found. Overall, most frequent adverse events, generally grade 1–2 in severity, included neutropenia (37.5% of patients), diarrhoea (37.5%), asthenia (33.3%), palmar-plantar erythrodysesthesia (25.0%), constipation (25.0%), pyrexia (16.7%), rash (12.5%), thrombocytopenia (12.5%) and anorexia (12.5%). Grade 3–4 adverse events were rare, and mostly resolved. Most frequent grade 3–4 AEs included neutropenia (20.8% of patients), palmar-plantar erythrodysesthesia (8.3%) and rash (8.3%). No treatment-related deaths due to toxicity occurred.

Conclusions: Continuous sorafenib 400 mg bid is generally well tolerated and showed preliminary anti-tumour activity in combination with DTIC in patients with advanced metastatic melanoma. This regimen warrants further evaluation in larger clinical trial settings.

792 CORRELATES OF OUTCOME IN PATIENTS WITH METASTATIC MELANOMA TREATED WITH DACARBAZINE

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Purpose: To identify correlates of overall and progression-free survival in patients with metastatic melanoma treated with dacarbazine chemotherapy.

Patients and methods: A retrospective single institution analysis was performed on a cohort of 109 patients with stage IV melanoma, who received chemotherapy with dacarbazine 750 to 1000mg/m² every three weeks [median 3 cycles (range 1 – 17)]. Multivariate proportional hazard models were used to examine the effects of key clinical indicators (age, gender, performance status (PS), primary site and stage, local/regional recurrence status, metastasis-free survival, sites of metastases, LDH, neutrophil levels and prior chemotherapy or immunotherapy).

Results: Median overall survival from the 1st metastasis was 12 months (95% C.I., 9.9 to 15.3 months); median progression-free survival was 2.3 months (95% C.I., 2.0 to 3.0 months). Median time to first metastasis was 32 months (95% C.I., 22 to 38 months). 57% of patients were still alive 6 months after starting dacarbazine chemotherapy. CR was observed in 7 (6.0%) patients and PR in 16 (15%). Of these 23 responders, the average duration of response was 13.4 months (95% C.I. 6 to 21 months). Presence of bone metastases (HR: 2.72), liver metastases (HR: 2.40), primary site in a non chronic sun-exposed area (HR: 2.5), and ECOG PS > 1 (HR: 1.5) were significant negative

correlates of overall survival. Similar results were obtained for progression-free survival. ECOG > 1 (OR = 2.8) was the only correlate of non-response to dacarbazine.

Conclusion: The presence of bone metastases, liver metastases, poor performance status and primary melanoma from non sun-exposed areas are independent negative correlates of overall survival and progression-free survival in patients with metastatic melanoma treated with dacarbazine. Performance status is the only clinical correlate with response.

793 PHASE I STUDY OF THE COMBINATION OF DOCETAXEL, TEMOZOLOMIDE AND CISPLATIN IN PATIENTS WITH METASTATIC MELANOMA

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Background. In a search for more effective combination chemotherapy for the treatment of metastatic melanoma, we conducted a phase I trial of a novel combination of docetaxel, temozolomide, and cisplatin. **Methods.** Patients with inoperable or recurrent metastatic melanoma with a Zubrod performance status (PS) of 2 or less and adequate organ function were eligible. The dose of docetaxel was escalated between cohorts of patients, and the doses of temozolomide and cisplatin were fixed. A standard 3+3 dose-escalation design was used to determine the maximum tolerated dose (MTD).

Results. Among 23 patients who were enrolled, 2 were evaluable for toxicity. The median age was 58 years, and the median Zubrod PS was . Eighteen patients (78%) had stage IV-Mc disease. The dose-limiting toxicities were myelosuppression and pulmonary embolism. The MTD was 3 mg/m² docetaxel on days, 8, and 5 when given with; 5 mg/m² temozolomide on days –5; and 2 mg/m² cisplatin on days –4, repeating every 4 weeks. Among 9 patients evaluated for response, 6 (32%) had partial responses and 5 (26%) had stable disease. Among 4 chemo-naïve patients, 6 (43%) had a partial response and 4 (29%) had stable disease. Nine patients developed brain metastases by the time of the last follow-up evaluation, and the median time to brain metastases for all 9 evaluable patients has not been reached.

Conclusion. This combination was well tolerated and appears to be a promising treatment for patient with metastatic melanoma. Further investigation with a phase II trial is warranted.

Table 1. Grade 3/4 Toxicities

	Level 0 (n=3)	Level 1 (n=3)	Level 2 (n=6)	Level 3 (n=6)	Level 4 (n=6)	Level 5 (n=6)	Level 6 (n=6)	Level 7 (n=6)
Toxicity	Gr3	Gr4	Gr3	Gr4	Gr3	Gr4	Gr3	Gr4
Neutropenia	0	0	2	1	1	1	2	1
Infection with neutropenia	0	0	0	0	0	0	0	1*
Lymphopenia	3	0	4	1	1	0	1	1
Anemia	0	0	1	0	1	0	0	0
Thrombocytopenia	0	0	1	1	1	0	1	0
Infection with normal neutrophil counts	0	0	0	0	1*	0	0	0
Vomiting	0	0	1	0	0	0	1	0
Diarrhea	0	0	1	0	0	0	0	0
Pulmonary embolism	0	0	0	1	0	1	0	1*
Fatigue	0	0	0	0	1	0	0	0

*Denotes the dose-limiting toxicity, as defined in the protocol, observed during the first cycle.

Table 2. Response to Treatment

Dose level	Total number	Partial response	Stable disease	Progressive disease	Not evaluable
0	3	1	0	2	0
1	6	1	2	3	0
2	6	3	1	1	1
3	8	1	2	2	3